

GPS Training Session Analysis

Athlete training session — 31 October 2018

Prepared for the coaching staff, by Luisa Molenda

Goal & Data

This report looks at one training session recorded with a 10 Hz GPS tracker (one position every 0.1 second) plus a heart rate monitor. The raw file has 59,551 rows and includes time, position (x/y in metres and GPS coordinates), speed and heart rate.

No information about the sport or the athlete was given. Based on the field size I was given (118 × 72 m) and the movement pattern, I assume this is a football session and I use football references for speed zones and sprint thresholds.

The goal is simple: turn the raw data into a clear picture of how hard the athlete worked, how they moved, and whether they showed signs of fatigue.

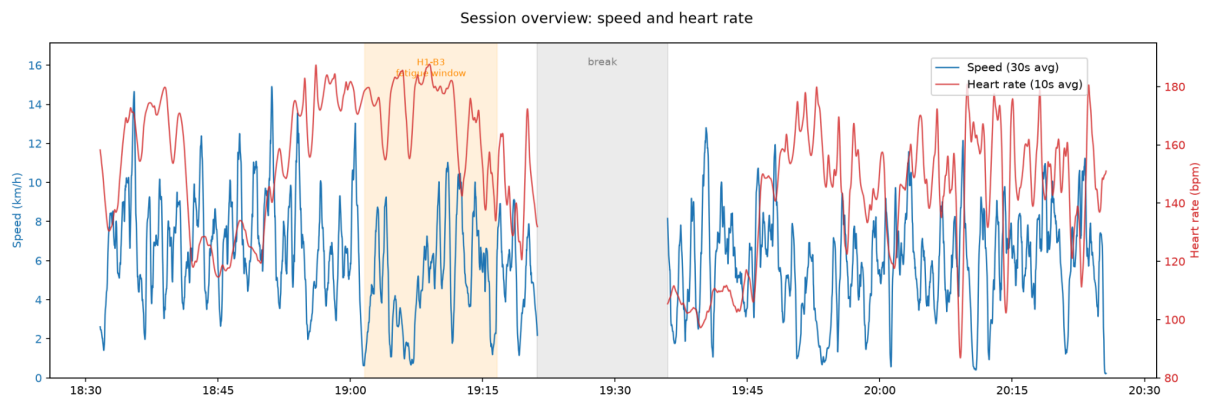
Data Validation and Preparation

Before any analysis, data quality was checked. The main points:

- **No missing values** in any column, and all values are in a normal range (speed 0–27 km/h, heart rate 86–189 bpm);
- **The recording is not fully continuous.** It spans 114 minutes, but there is one gap of about 15 minutes (19:21 to 19:36). This splits the session into two halves of about 50 minutes each. I treat this gap as a half-time break and never calculate distance or acceleration across it;
- **Heart rate signal is clean:** no stuck or frozen values;
- **Speed was cross-checked:** I calculated speed a second way, from the position data, and compared it with the device speed. They match closely (correlation 0.94, no bias), so the device speed is reliable. I use it for all calculations;
- A few points (16 seconds total) fall just outside the field line. They all happen right before and after the break. The assumption made was that the athlete was walking off and back on;

Full method and threshold details are in the appendix.

Session Overview



The athlete covered 9.96 km in about 99 minutes of active time, across two halves. Heart rate averaged 149 bpm, which shows a demanding session. The chart above shows the shape of the session: effort and heart rate climb through the first half, drop during the break, and stay lower and steadier in the second half.

Key numbers:

Metric	Values
Total distance	9,960 m (H1: 5,137 / H2: 4,823)
Distance above 20 km/h	465 m (H1: 292 / H2: 173)
Max speed	27.32 km/h (at 20:23:08.8)
Sprints ≥ 25 km/h	4
Sprints ≥ 21 km/h	18 (H1: 9 / H2: 9)
Hard accelerations (>3 m/s ²)	12
Hard decelerations (<-3 m/s ²)	31
Average heart rate	149 bpm (H1: 157 / H2: 141)
Max heart rate	189 bpm
Time above 90% max HR	23.6 min

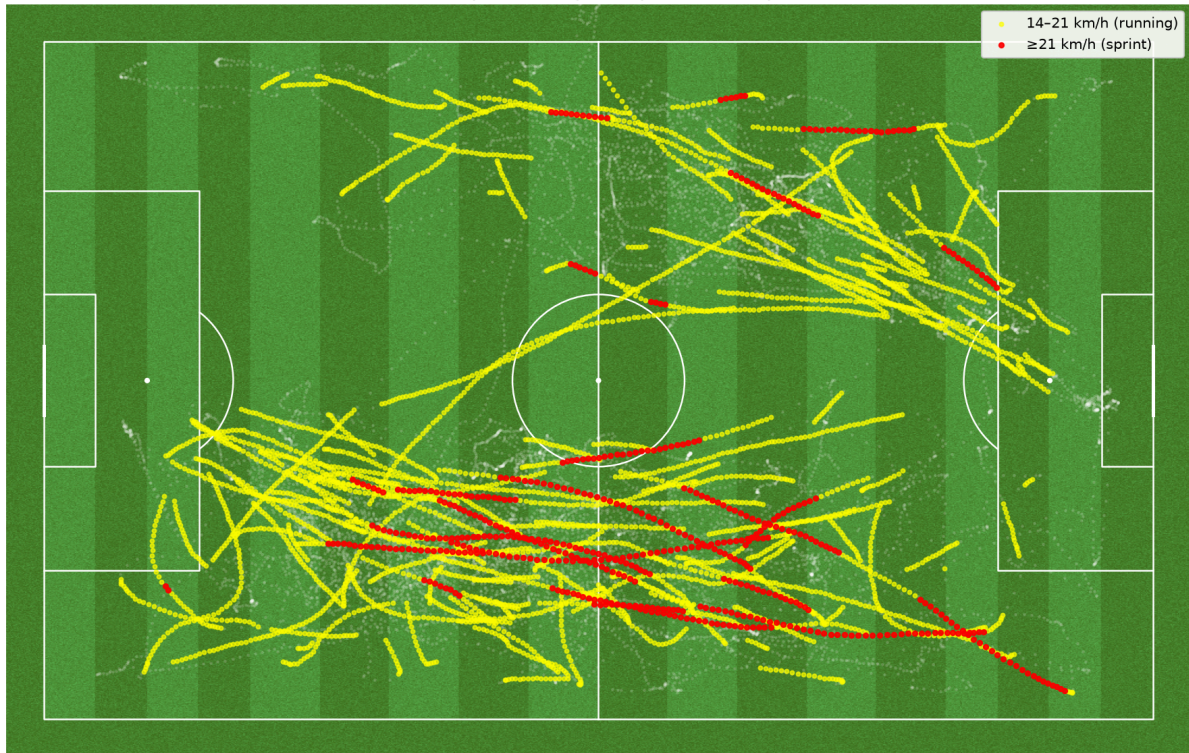
One clear point for training design: the athlete brakes hard much more often than they accelerate hard (31 vs 12). Hard braking is a main cause of muscle damage, so this is useful to know for recovery planning.

Athlete Movement (Where and How)

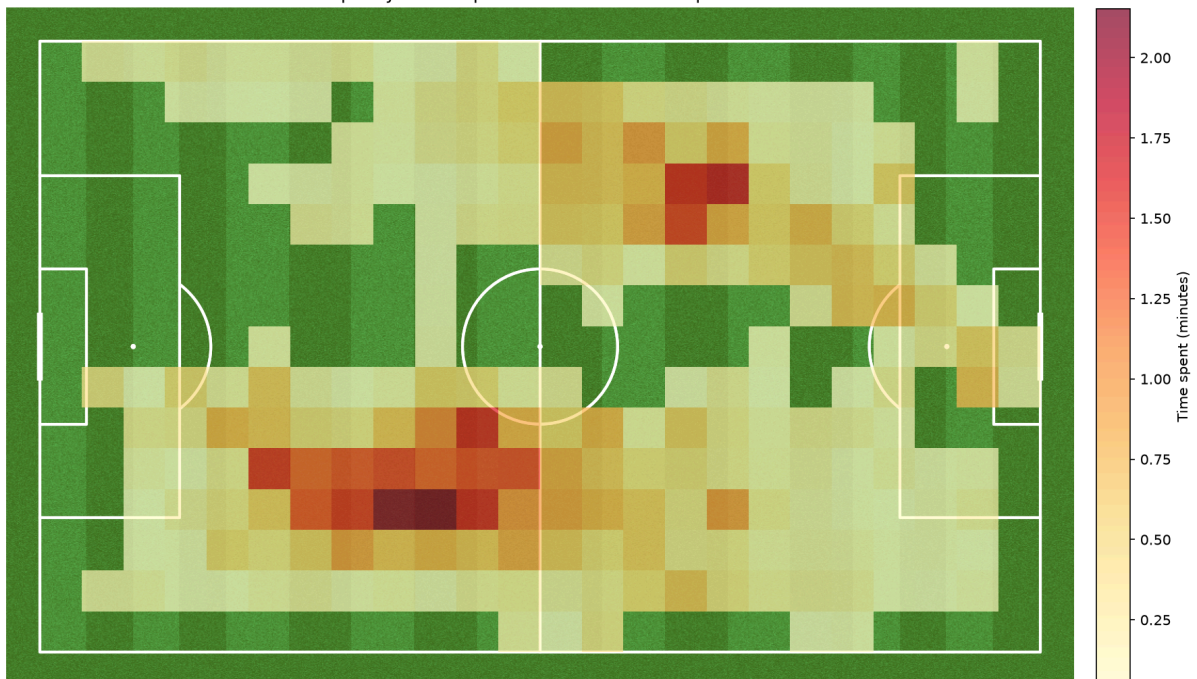
The movement map shows running (yellow) and sprints (red) by location. High-speed running spreads across the pitch in long diagonal runs, but the true sprints are more

concentrated in the lower channel, the fast efforts happen mostly in one lateral zone. The heatmap shows where the athlete spent time. There are two main zones on opposite diagonals, around the centre of the pitch. This could mean a role that covers two areas or simply two different drills in two areas.

Movement map — running and sprint efforts by location

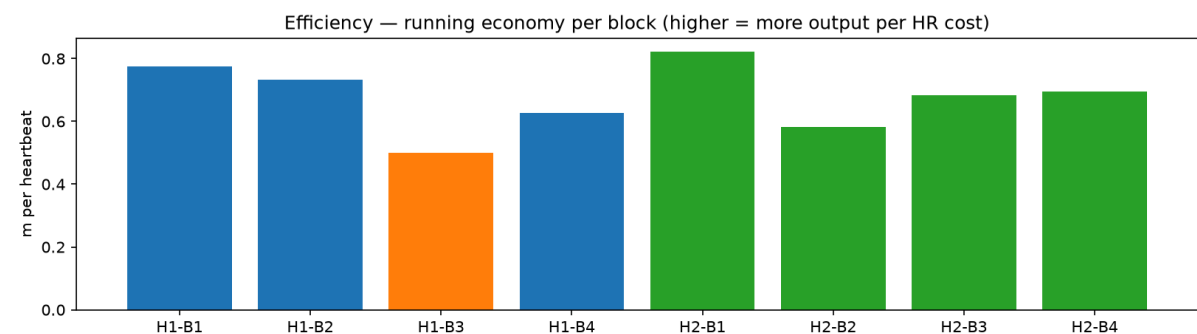
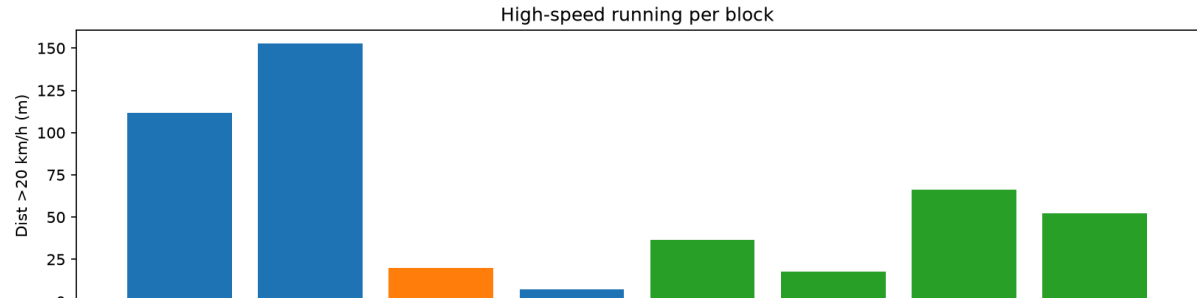
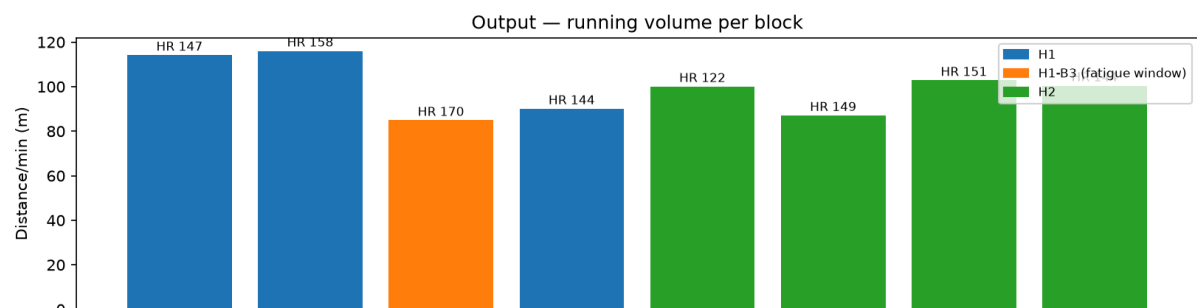
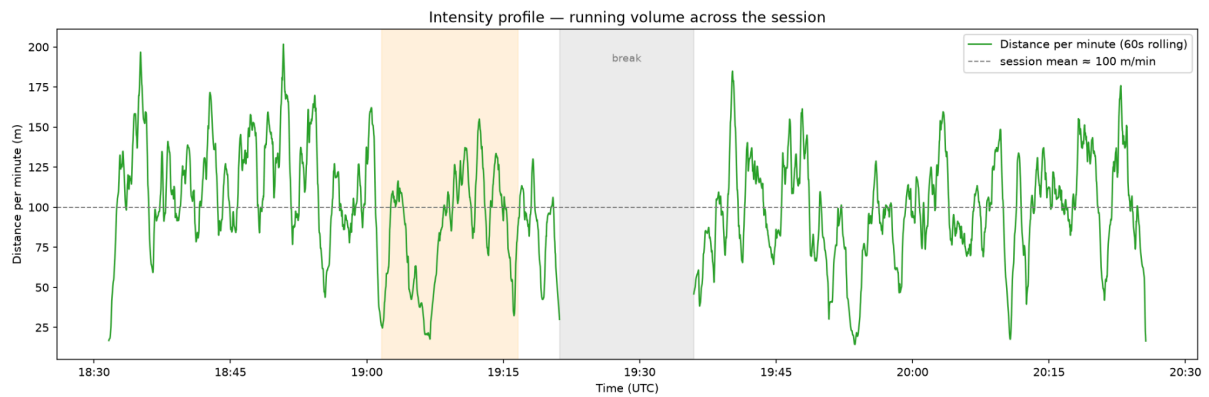


Occupancy heatmap — where the athlete spent time



Fatigue During The Session

Each half was split into 15-minute blocks and tracked output, high-speed running and an efficiency index (metres covered per heartbeat, higher means more running for less heart cost).



The pattern is clear:

- In the first 30 minutes, the athlete produced high output and increasing high-speed running (112 m, then 153 m per block above 20 km/h);
- Then, in the block from 19:01 to 19:16, everything drops at once: output falls to its lowest, high-speed running almost stops (20 m) and heart rate is at its highest (170

bpm). The athlete is doing less while the body works harder, a sign of fatigue or a very high-cost effort;

- High-speed running never returns to first-half levels after this point. The best second-half block reaches only 66 m above 20 km/h, against 153 m in the first half;

The half-time break lowered the heart rate well (the restart block is low and easy), but it did not bring back the earlier running economy. In the second half, at similar heart rates, the athlete produced less running than in the first half.

Coach Takeaways

- **The athlete front-loads high-speed work:** Most fast running happens in the first 30 minutes. If the goal is to build repeated sprint ability, this early peak and the sharp drop after suggest the athlete may not sustain high-speed output across a full session yet;
- **The 19:01–19:16 block is the key moment:** Whatever happened there (fatigue or a hard drill), it marks the point where high-speed running stopped for the rest of the session. Worth checking against the session plan;
- **High braking load:** The athlete decelerates hard often. Include this in recovery planning, since hard braking drives muscle soreness;
- **Peak speed stayed available late.** The fastest sprint of the whole session came near the end (20:23). So maximal speed was still there, what dropped was the *repeated* high-speed running, not top-end capacity. This is a useful distinction: the issue looks like endurance of high-speed work, not raw speed;

Appendix (Method & Assumptions)

- **Speed in m/s:** $\text{km/h} \div 3.6$. **Distance:** $\text{speed} \times \text{time step (0.1 s)}$, never across the break. Cross-checked with position-based distance (within 1%);
- **Smoothing:** speed smoothed with a 0.5-second moving average before calculating acceleration, to avoid false spikes from GPS noise. Acceleration values stay within a normal human range (about $\pm 4.8 \text{ m/s}^2$);
- **Speed zones (km/h):** standing <1, walking 1–7, jogging 7–14, running 14–20, high-speed 20–25, sprinting >25 (football references);
- **Sprint definition:** sprints were defined at 21 km/h, not the 25 km/h. The reason is simple, this athlete rarely goes above 25 km/h (only 4 times in the whole session), so a map at 25 km/h would show almost nothing. At 21 km/h there are 18 efforts, which gives a clearer picture of where the fast running happens.
- **Heart rate zones:** as % of the session maximum (189 bpm), used as a proxy because the true maximum heart rate is unknown/
- **Efficiency index:** $\text{distance per minute} \div \text{average heart rate}$, per block. A simple measure; it is affected by heart rate lag and drift, so I use it as support, not proof;
- **Main limitation:** no data on the athlete (age, sex, level) or the session plan. Some findings (role, fatigue vs drill design) cannot be fully confirmed without this;
- **Code & Data:** The full analysis code (Jupyter notebook) is available on GitHub repo: [data-analytics-jci](#). The raw data is not included in the repository, due to its size, but still shareable. The dataset is available at [\[Data Analytics in Sports JCI\] Assignment #3](#).